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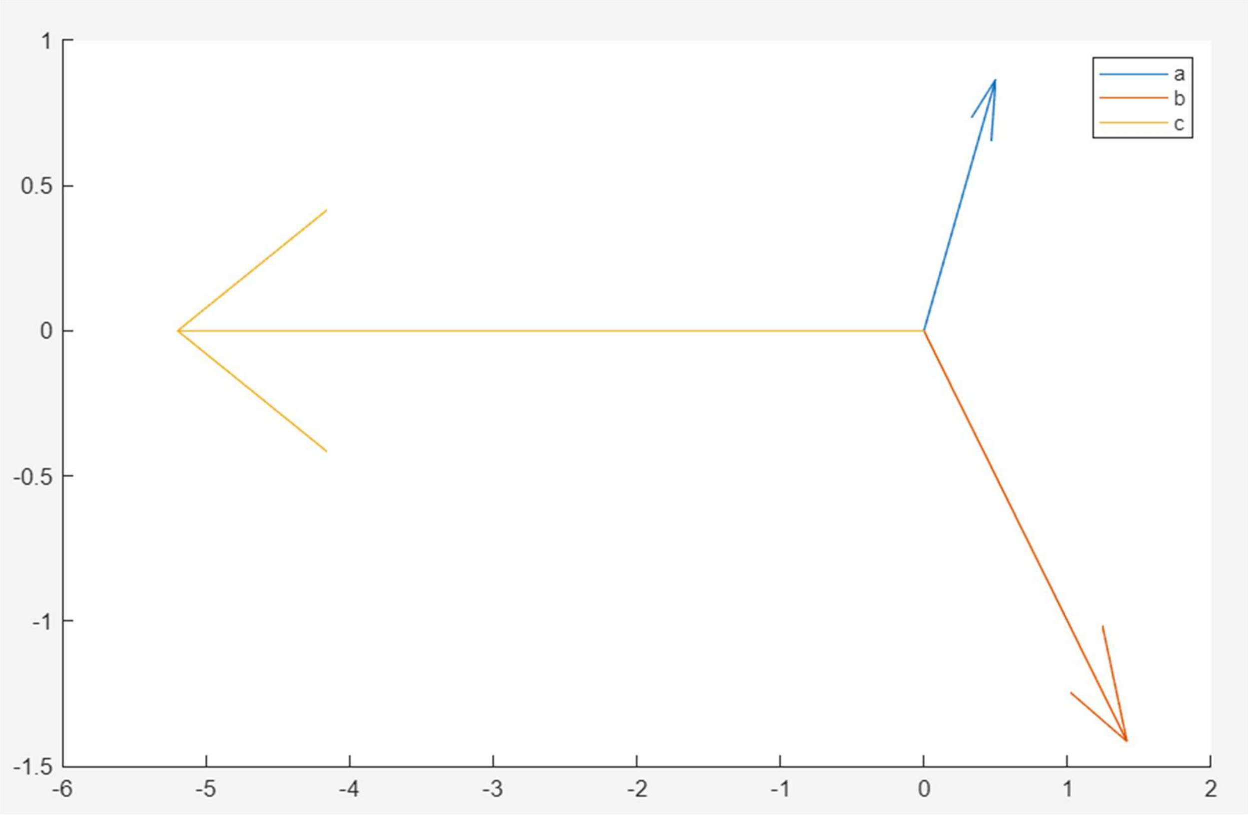
EEL 3135 – Classwork #03

Milestone:

- Question 2 must be completed and choose any other question you would like.

Question #1: Euler's Formula, express the following signals in both rectangular and polar form. Graph parts a,b,c and their phasor representation on the complex plane on Matlab.



1. $e^{\frac{j\pi}{3}}$
   a. $= \cos\left(\frac{\pi}{3}\right) + jsin\left(\frac{\pi}{3}\right) = \frac{1}{2} + j\left(\frac{\sqrt{3}}{2}\right)$
   b. $r = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} = 1$
   c. $\phi = \arctan\left(\frac{\left(\frac{\sqrt{3}}{2}\right)}{\frac{1}{2}}\right) = \frac{\pi}{3}$
   d. $1\angle 60$

2. $2e^{-\frac{j\pi}{4}}$
   a. $= 2\left[\cos\left(-\frac{\pi}{4}\right) + jsin\left(-\frac{\pi}{4}\right)\right]$
   b. $= 2\left[\left(\frac{\sqrt{2}}{2} - j\frac{\sqrt{2}}{2}\right)\right]$
   c. $= \sqrt{2} - j\sqrt{2}$
   d. $r = 2$
   e. $2e^{-\frac{j\pi}{4}} = 2\angle -\frac{\pi}{4}$
3. $3e^{j\left(\frac{5\pi}{6}\right)} + 3e^{-j\left(\frac{5\pi}{6}\right)}$
   a. $= 6\cos\left(\frac{5\pi}{6}\right)$
   b. $= -5.196$
   c. $= 5.196\angle 180$
4. $\cos(200\pi t + \pi/4)$
   a. $= \frac{1}{2}\left[e^{j\left(200\pi t+\frac{\pi}{4}\right)} + e^{-j\left(200\pi t \quad \frac{\pi}{4}\right)}\right] \rightarrow\ A = 1; \omega = 200\pi; \phi = \frac{\pi}{4}$
   b. $1\angle\frac{\pi}{4}$

Question #2: The two complex sinusoids:

$x_1(t) = 3.5\cos(120\pi t + \pi/5)$, $x_2(t) = 2.5\cos(120\pi t + 2\pi/3)$.

(a) Write $x_1(t)$ and $x_2(t)$ in exponential form using Euler's identity.

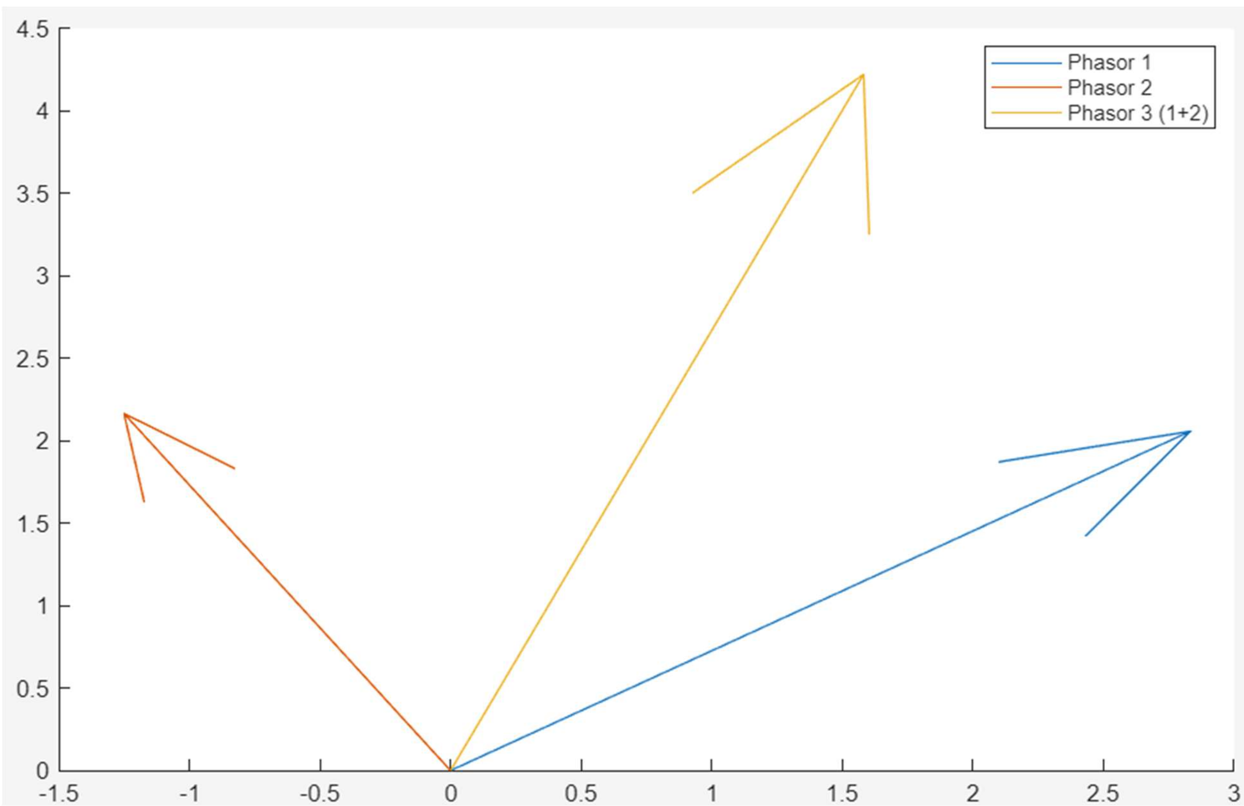
$x_1(t) = 1.75e^{\left(j*120\pi t+\frac{\pi}{5}\right)} + 1.75e^{\left(-j*120\pi t+\frac{\pi}{5}\right)}$ | $x\_2\ (t) = 1.25e^{\left(j*120\pi t+\frac{2\pi}{3}\right)} + 1.25e^{\left(-j*120\pi t\ \ \frac{2\pi}{3}\right)}$

(b) Represent each signal as a phasor ($Ae^{j\phi}$).

$$x_1(t) = \ 3.5e^{j\left(\frac{\pi}{5}\right)}\ |\ x_2(t) = 2.5e^{j\left(\frac{2\pi}{3}\right)}$$

(c) Add the two phasors graphically and determine the resulting amplitude and phase (do this in MATLAB).

$$4.5088e^{j*1.2124}$$



(d) Express $x_1(t) + x_2(t)$ in the form $A\cos(\omega t + \phi)$

$$4.5088\cos\left(120\pi t\ +\ 1.2124\right)$$

Question #3: Given

$$X(t) = 6 + 12\cos(300\pi t - \pi/6) + 5\cos(700\pi t - 2\pi/3),$$

(a) Determine the fundamental frequency of $X(t)$.

(b) Write $X(t)$ in terms of complex exponentials, identifying amplitude and phase of each frequency bin.

(c) Write MATLAB code to generate the magnitude and phase spectrum for the signal. Fill in the missing parameters:

```
clear all; close all; clc;
% Define waveform components
A1 = ___; f1 = ___; p1 = ___;
A2 = ___; f2 = ___; p2 = ___;
% Fundamental frequency f =
gcd(f1,f2); f_bin = -___:f:___;
% Construct phasors
A = [ ... ];
Phi = [ ... ];
Phasors = A .* exp(1j*Phi);
% Plot spectra figure; stem(f_bin, abs(Phasors));
title('Magnitude Spectrum');
figure; stem(f_bin, angle(Phasors)); title('Phase Spectrum');
```

(d) Describe the **magnitude spectrum** and **phase spectrum**.

Question #4: Consider the following composite signal:

$$X(t) = 8 + 10\cos(320\pi t + \pi/3) + 7\cos(560\pi t + 3\pi/8).$$

(a) Use Inverse Euler's Formula to expand $X(t)$ into complex exponentials.

(b) Determine the fundamental frequency of the signal and express the spectrum in terms of harmonics.

(c) Use Matlab to graph both the magnitude spectrum and phase spectrum.

(d) Identify the DC offset and explain its role in the spectrum.